Safe Welding, Cutting, and Other Hot Work Practices in Refineries, Gas Plants, and Petrochemical Plants

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STEP
Strategies for Today's Environmental Partnership

American Petroleum Institute
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Health and Environmental Affairs Department
Safety and Fire Protection Group

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FOREWORD

This publication provides guidelines for the protection of personnel and property when welding, cutting or other hot work is performed in and around refineries, gas plants, or petrochemical plants.

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Safe Welding, Cutting, and Other Hot Work Practices in Refineries, Gas Plants, and Petrochemical Plants

SECTION 1—GENERAL

1.1 Scope

This publication provides guidelines for safe practices when welding, cutting or other hot work is performed in refineries, gas plants, or petrochemical plants.

1.2 Definition of Terms

Terms used in this publication are defined in 1.2.1 through 1.2.6. See 1.3 for publication data.

1.2.1 A *competent person* means one who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to personnel, and who has authorization to take prompt corrective measures to eliminate them. [This definition is from OSHA, 29 *Code of Federal Regulations* (CFR) 1926.32.]

1.2.2 *The lower flammable limit (LFL)* is the minimum concentration of a vapor in air (or other oxidant) below which propagation of flame does not occur on contact with an ignition source. The lower flammable limit is usually expressed as a volume percentage of the vapor in air. [This definition is based on the definitions of flammable (explosive) limits given in NFPA 325M.]

1.2.3 *The upper flammable limit (UFL)* is the maximum concentration of a vapor in air (or other oxidant) above which propagation of flame does not occur on contact with an ignition source. The upper flammable limit is usually expressed as a volume percentage of the vapor in air. [This definition is based on the definitions of flammable (explosive) limits given in NFPA 325M.]

Note: In popular terms, a mixture above the UFL is too "rich" to burn and one below the LFL is too "lean" to burn.

1.2.4 An *exposure limit* is a measure of the maximum airborne concentration limits for toxic substances to which workers may be exposed without protection (for example, respirators). Exposure limits are usually expressed in parts per million or milligrams per cubic meter. Material Safety Data Sheets (MSDSs) from the manufacturer or supplier of the material should list exposure limits as established by OSHA or other organizations (See 1.2.4.1 and 1.2.4.2).

1.2.4.1 *Permissible exposure limits (PELs)* are federal regulations set by the Occupational Safety and Health Administration, U.S. Department of Labor, and found at 29 *Code of Federal Regulations* 1910.1000, and in the substance-specific standards which follow. The latest editions should be consulted.

1.2.4.2 *Threshold limit values (TLVs)*[1] are recommended exposure limits published annually by the American Conference of Governmental Industrial Hygienists in the booklet *Threshold Limit Values and Biological Exposure Indices.* The latest edition should be consulted.

1.2.5 A *confined space* is an enclosure with known or potential hazards and restricted means of entrance and exit, is not normally occupied by people, and is usually not well ventilated. Examples of confined spaces in the petroleum industry include the following: process vessels and related equipment (vessel tower skirts and flare stacks), boilers, storage tanks, tank cars and trucks, vaults, large-diameter piping, and under certain circumstances, spaces located below ground level, such as pits (See OSHA definition from 29 CFR 1910.146).

1.2.6 *Hot work* is an operation that can produce a spark or flame or other source of ignition having sufficient energy to cause ignition, where the potential for flammable vapors, gases, or dust exists. Hot work includes such things as arc, gas and cad welding, chipping, flaming, grinding, gas cutting, sandblasting, and soldering.

1.3 Referenced Publications

The latest editions of the following publications are referenced in this publication:

ACGIH[2]

Threshold Limit Values and Biological Exposure Indices

ANSI[3]

Z49.1 *Safety in Welding and Cutting*

API

Std 590 *Steel Line Blanks*

Publ 570 *Piping Inspection Code Inspection, Repair, Alteration, and Re-Rating of In-Service Piping Systems*

Std 2015 *Cleaning Petroleum Storage Tanks*

RP 2201 *Procedures for Welding or Hot Tapping on Equipment in Service*

Publ 2202 *Dismantling and Disposing of Steel From Tanks Which Have Contained Leaded Gasoline*

[1]TLVs is a registered trademark of the American Conference of Governmental Industrial Hygienists.

[2]American Conference of Governmental Industrial Hygienists, 6500 Glenway Avenue, Building D-7, Cincinnati, Ohio 45211-4438.

[3]American National Standards Institute, 1430 Broadway, New York, New York 10018.

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Publ 2217A *Guidelines for Work in Inert Confined Space in the Petroleum Industry*

BMUSDI[4]

Flammability Characteristics of Combustible Gases and Vapors, Bulletin 627

NFPA[5]

10 *Portable Fire Extinguishers*

325M *Fire Hazard Properties of Flammable Liquids, Gases, and Volatile Solids*

OSHA[6]

29 *Code of Federal Regulations*, Sections:

1910.119 "Process Safety Management of Highly Hazardous Chemicals"

1910.146 "Permit-Required Confined Spaces"

1910.147 "The Control of Hazardous Energy (Lockout/Tagout)"

1910.251 "Welding, Cutting, and Brazing"

1910.1000 "Air Contaminants"

1926.32 "Definitions"

1926.350 "Gas Welding and Cutting"

1926.354 "Welding, Cutting, and Heating in Way of Preservative Coatings"

1.4 Additional Reference Material

The latest editions of the following are related reference materials:

API

Std 2610 *Design, Construction, Operation, Maintenance, and Inspection of Terminal and Tank Facilities*

NFPA

30 *Flammable and Combustible Liquids Code*

51B *Cutting and Welding Processes*

1.5 General Precautions

Gas and electric cutting and welding operations are important adjuncts to petroleum operations and require a high degree of skill. Personnel engaged in these operations should have a thorough understanding of the duties they are to perform and the potential hazards to guard against. These personnel shall be qualified and properly instructed to ensure that safe work procedures are followed. In all welding and cutting operations, the welder and other personnel related to the job shall be briefed on the health and safety aspects of their specific jobs before any work is started, so that they will follow safe work practices and facility procedures, including emergency response.

SECTION 2—JOB ANALYSIS AND SPECIAL PRECAUTIONS

2.1 Job Analysis

Permits should be used to authorize and control all hot work. To determine the appropriate safeguards and required personal protective equipment, a careful analysis should be made of the potential hazards and the operations to be performed prior to starting work. This safety analysis can be part of the normal permit procedure. Some of the factors that should be considered in the pre-job safety planning are described in 2.2 through 2.5.

2.2 General

Precautions shall be taken to prevent uncontrolled ignition of flammable or combustible materials. With proper procedures, restricting access to the work area, issuance and posting of detailed permits as appropriate, and adherence to site-specific safe work conditions, hot work can be safely conducted. Normally, the work area should be hydrocarbon vapor and gas free. Under permit conditions, however, hot work may be performed on equipment in service with hydrocarbon vapors or gases present, provided that (a) hot work is performed in an oxygen deficient atmosphere, such as in an inerted space, or (b) hydrocarbon vapors or gases are controlled so as to remain within a predetermined percent of the LFL, or (c) the equipment is in a well-ventilated area, and precautions have been taken to ensure there is no accumulation of hydrocarbon vapors or flammable gases to create an explosive atmosphere or major fire hazard. In addition, precautions should be taken to prevent burn-through (see API RP 2201).

Before welding or cutting is performed above or near oily surfaces, the area should be flushed with water, steam cleaned, or covered with clean dirt or sand, or other precautions should be taken. Holes in the ground or cracks in pavement should be similarly flushed or covered. All sewer manholes and catch basins in the vicinity of the work area should be covered to prevent hydrocarbon vapors or gases from exiting the sewer or drain and reaching an area of hot work, or to prevent sparks from getting into the sewer. If a catch basin is equipped with a trap, the basin may be continuously flushed with water. If sewer manholes or drains are covered in preparation for hot work, the covers must be

[4]Bureau of Mines, U.S. Department of the Interior, *Flammability Characteristics of Combustible Gases and Vapors* by Michael G. Zebetakis, Bulletin 627, 1965, is available through the Bureau of Mines, Washington, D.C.

[5]National Fire Protection Association, Batterymarch Park, Quincy, Massachusetts 02269-9990.

[6]Occupational Safety and Health Administration, U.S. Department of Labor. *The Code of Federal Regulations* (CFR) is available from the U.S. Government Printing Office, Washington, D.C. 20402.

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removed when the work has been completed or at the end of the day or shift (if the work is not completed) and replaced when work starts again. If wood scaffolding or other combustibles are in use, appropriate precautions should be taken to protect them against ignition.

2.3 Wind

Hot work should not be permitted in areas into which flammable vapors could be blown, for example, from leaking process piping or equipment in adjacent areas or in the event of a pressure-relief valve discharging into the atmosphere. Wind can also carry sparks and embers from the hot work into adjoining process areas and ignite any vapors leaking from flanges, valve packings, sewer catch basins, vents, gauging hatches, and so forth (see 2.6).

2.4 Flammable Liquids, Vapors, Solids, or Dusts

Hot work should not be permitted while adjacent equipment that contains flammable liquid, vapor, solids, or dust is being opened, disassembled, steamed, ventilated, or flushed of sediment without considering how such actions might affect the hot work, or visa versa. Attention should be given to drums or other portable containers which contain flammable or combustible materials. Hot work performed in areas where tanks are receiving flammable liquids or gases should be continuously monitored to ensure that the atmosphere is safe. In areas where hot work is approved, operators must not release flammable liquids or vapor until the hot work has stopped.

2.5 Hot Work in Confined Spaces

If hot work requires entry into a confined space, the permit system should address both the hazards of entry and of doing hot work in a confined space. Also, OSHA-permit required confined-space procedures may apply (see API Publ 2217A and OSHA, 29 CFR 1910.146). With the exception of inert spaces, the concentration of flammable vapors present must always be less than 10 percent of the LFL, regardless of the type of respiratory protection required in order to permit entry. Note the following:

a. Special precautions must be taken in the event that oxygen levels increase above 23.5 percent, as this affects the flammable limits of any materials present, and the LFL will be lowered.

b. Provisions should also be made to ensure that adequate ventilation is provided to each welder and that cutting or welding operations inside a confined space do not create a hazard to personnel (see OSHA, 29 CFR 1910.252 and ANSI Z49.1). For example, burning could deplete the oxygen in a confined space, and welding or cutting exotic metals or surfaces which are galvanized or lead contaminated could create a toxic atmosphere. Also, leaving an unattended oxygen/acetylene hose attached to a torch in a confined space could increase oxygen or flammable gas concentrations as a result of a leak.

c. Care should be taken to ensure that exhaust fumes from internal combustion engines, such as welding machines, generators, or vehicles, do not contaminate confined spaces.

If air-supplied or self-contained breathing apparatus (SCBA) for respiratory protection is not required, precautions must be taken to ensure that the atmosphere will at all times meet the following conditions when personnel are present:

a. An oxygen content, by volume, of at least 19.5 percent and no more than 23.5 percent.

b. Concentrations of toxic vapors and gases below the established exposure limit (see 1.2.4 and Section 8).

2.6 Fire Watch Personnel

Properly trained and equipped fire watch personnel shall be required whenever hot work is performed in locations where flammable vapors or gas may be present, where fire might develop from combustible liquids or Class A (as defined in NFPA 10) materials, or when special conditions may warrant the use of fire watch personnel. The person assigned to a fire watch shall have no other duties that would distract from the primary safety function. One should consider having fire watcher personnel wear special vests or clothing so that they can be easily recognized.

Fire watch personnel shall have the appropriate fire extinguishing equipment readily available and shall be trained in the equipments' use. They shall be familiar with facilities for sounding an alarm in the event of a fire. They shall watch for fires in all exposed areas, try to extinguish them only when obviously within the capacity of the equipment available, or otherwise sound the alarm. After completion of welding/cutting, the fire watch shall be maintained until the area has been inspected and found to be free of fires or smoldering materials.

2.7 Welding Machines

Welding machines can be an ignition source and must be treated as such. Care must be taken to ensure that welding machines and leads are properly grounded to avoid stray currents, and that the welding machine and the grounding connection are located in an area free of flammable vapors or gases. Welding machines should be inspected before use to ensure that there are no frayed or broken wires, all connections are tight, and the machine functions properly. Welding leads should be protected from chaffing when passing through openings, across access ways, and so forth.

2.8 Equipment With Liners

Careful inspections must be performed and special precautions taken on any vessels, tanks, piping, or other

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equipment which is lined and which could have void spaces (for example, vessels with stainless steel liners) to ensure that no product or vapors are trapped between the layers. This includes tanks with double bottoms, fiberglass-lined tanks, double-walled tanks and piping, and the like.

SECTION 3—HOT WORK PERMITS

3.1 General

A written hot work permit system should be used to control all hot work within units used to process or store flammable or combustible materials. Authorization of those hot work permits should be vested in the personnel in charge of operating such process or storage equipment. Operations personnel should immediately notify workers engaged in hot work if actual or imminent changes in conditions (for example, a release of material from a pressure-relief valve) may endanger their safety so that the hot work can be stopped at once.

3.2 Permit Requirements

Except in areas specifically designated as safe for hot work, such as maintenance welding shops or outlying fabricating areas, a permit shall be obtained before starting any work that can involve a source of ignition. (Equipment such as vessels and piping should be gas free and cleared of toxic substances and liquid hydrocarbons before being brought into weld shops or the designated safe areas.) Each permit for hot work should state precautions necessary for the specific job. The permit shall be in writing and should include the following information and conditions:

a. The equipment to be worked on and the location of work.
b. The type of work to be performed (for example, cutting, welding, grinding).
c. The protective equipment and protective measures required, including rescue equipment for work in confined spaces.
d. Standby fire protection equipment and/or personnel required.
e. The permit's expiration date and time.
f. The area/equipment required to be tested with an oxygen and combustible gas detector and, if required, tested for toxics, plus the test results.
g. Whether continuous monitoring is required.
h. Any special precautions which are necessary to complete the job safely, including conditions for stopping work.
I. Indication that affected personnel have been notified.
j. Signatures of authorized individuals, that is, at least the person issuing the permit and the permit recipient (who in some instances may be a contract worker).
k. The time and date the permit was issued.

If hot work is to be performed in a confined space, the confined space permit should be supplemented or combined with a hot work permit.

Once the safe work procedures and permit requirements and conditions have been determined, and before the permit is issued, a careful analysis and inspection shall be made, including oxygen and flammable vapor tests, to determine that unconfined flammable vapors or gases or combustible materials are not present in the work area or equipment and that no reasonable probability exists of any such material entering the area while hot work is performed. The authorizing individual, who must be a competent person, may then issue a permit for the work to proceed. If several work groups are involved in the work process, the permit system should ensure adequate communication. A copy of the permit shall be kept on site until the work is completed. OSHA standards (29 CFR 1910.147) address specific requirements for written procedures associated with locking out and tagging out equipment and other energy sources. The permit process should indicate who is responsible for inspecting the work site, conducting gas testing, and approving the permit allowing hot work to proceed.

CAUTION: The fact that a permit has been issued for the performance of certain work in no way relieves those doing the work of their responsibilities for the safe execution of the assigned task in accordance with the permit requirements. If any unsafe condition arises, work should be stopped and the condition reported immediately.

3.3 Extending the Permit

When it is necessary to continue the work beyond the designated time, the authorizing individual on each succeeding day or shift shall ensure that conditions are satisfactory for work to continue and may then reissue the permit; if conditions have changed, it may be necessary to issue a new permit or amend the existing permit.

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SECTION 4—TESTING FOR FLAMMABLE VAPOR AND OXYGEN

A competent person using an appropriate combustible-gas detector should perform tests to determine flammable vapor concentrations before hot work is started. The combustible gas detector must be properly maintained, adjusted, and calibrated. The surrounding area and equipment and containers that might contain flammable vapors must be thoroughly checked, even if such equipment has been steamed, flushed, or otherwise cleaned. Most hydrocarbon vapors are heavier than air. Gas testing should include all low points such as sumps, drains, liquids boots, and so on, and all confined areas, such as floating roof pontoons on aboveground storage tanks, piping, vessels, and the like. Long runs of piping should be tested at multiple locations.

A combustible-gas detector may not detect the presence of oils with flash points substantially above ambient temperatures, such as those sometimes found in catalyst beds during unit shutdowns. Such oils that may be heated during the hot work process constitute a hazard and must be removed before cutting or welding is started. Detector tests should be conducted only after steaming operations have been completed so that steam vapors will no longer affect detector results. Nitrogen, argon, and hot gases may affect some types of combustible-gas instruments, possibly resulting in inaccurate readings. Anytime a tank, vessel, or piping has been purged with or has contained an inert gas (for example, nitrogen, argon), or anytime the possibility exists for an oxygen-enriched or oxygen-deficient atmosphere above or below the amount required by the instrument manufacturer to give a correct reading—an oxygen reading must be taken using a properly maintained and adjusted oxygen meter before the flammable vapor test. Consult the instrument manufacturer's literature for further information.

SECTION 5—PROTECTION OF PERSONNEL

Appropriate steps must be taken during hot work operations to protect personnel from potential and actual safety and health hazards such as the following:

a. Skin burns from sparks, slag, and the like.
b. Ultraviolet light rays from electric welding.
c. Welding fumes. The toxicity of welding fumes depends on their composition and concentration. The composition and quantity of fumes are determined by the materials being welded, the composition of the rods and any coatings or paints, the process being used, and the circumstances of use. Toxic fumes are generated from welding on metals that are coated with or contain alloys of lead, zinc, cadmium, beryllium, chromium, and certain other metals. Some paints may produce toxic fumes when heated. The potential health effects vary in type and severity depending on many factors, and some effects can be extremely serious.
d. Other toxic substances. Atmospheric testing and monitoring, along with personal protective equipment or other means to keep personnel exposures below OSHA Permissible Exposure Limits and/or Threshold Limit Values, should be implemented as appropriate. Note: A flammable vapor or gas concentration less than 10 percent LFL provides no assurance that exposures to certain materials will be less than their PELs (for example, benzene, lead, total hydrocarbons). Specialized monitoring and testing for these types of materials are required if possibilities for exposures above the PELs exist.
e. Flying particles during chipping and slag removal.
f. Oxygen deficiency in poorly ventilated spaces.
g. Setting combustible materials, oil-soaked insulation or workers' outer clothing on fire from sparks or slag.
f. Accumulation of flammable vapor or gas.
g. Oxygen or gas from welding equipment leaking into confined spaces.
h. Oxygen content rising to unsafe levels in inerted confined spaces.
i. Toxic preservatives which should be stripped away (see OSHA 29 CFR 1926.354).

SECTION 6—WORK ON VESSELS, EXCHANGERS, AND TANKS

6.1 Preparations

A confined-space entry permit, indicating the requirements and conditions for entry, shall be issued before any work is done inside a confined space. Before any work is done inside a vessel, exchanger, or tank, the space should be completely isolated from energy and potential energy sources by blinding, blanking and/or disconnecting all lines connected to the space, or by other approved means. A connection is considered properly blinded when the correct blind, gaskets, and bolts are properly installed to ensure that no leakage occurs. The open ends of connecting piping should be properly blinded, and the vessel should be thoroughly cleaned (unless inerted). [Additional special precautions are necessary if the vessel has a nonmetallic liner, has an internal coating, or has been used in leaded gasoline service or

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other toxic material service (see API Std 2015 and Publ 2202).] Energized equipment should be locked and/or tagged out. The vessel should be gas freed, tested for oxygen, then tested for combustible gas and/or toxics (unless inerted). Appropriate written procedures should be developed and used when conducting such operations.

6.2 Monitoring and Purging of the Atmosphere

Periodic combustible gas and oxygen retests, or continuous monitoring, may be required while hot work is proceeding. If the detector registers an excess or lack of oxygen or the presence of vapors in excess of that allowed by the permit, work should be stopped immediately, and the source of vapor or oxygen should be located and controlled (see API Std 2015).

For situations where the work in a vessel that has previously been pronounced gas free is delayed or suspended, the permit system shall specify the length of time beyond which detector tests must be repeated.

Before entry and during work, the atmosphere inside the vessel (if not inerted) should be continuously purged—using an appropriate air exchanger—if natural ventilation is inadequate.

6.3 Inspection of Internals

An inspection should be made to ensure that oil, residue, vapor, and other flammable and combustible materials have been removed or protected from ignition by sparks or pyrophoric iron sulfide. This includes areas such as bubble-cap trays, weirs, internal pans, vapor seals, double bottoms, pontoons, linings, under floors, or roof drain piping. If any doubt exists about the cleanliness of internal equipment, the equipment should be removed, cleaned, or suitably protected.

6.4 Work on Outside Surfaces

When hot work is limited to the outside surface of an empty vessel that has been isolated and gas freed, internal surfaces in the vicinity of the hot work which could become heated should be cleaned of any oil, wax, or pyrophoric iron sulfide deposits, prior to starting hot work. In all cases where hot work is being done on the outside surface of a vessel, adequate provisions, such as venting, should be made to protect against overpressure due to thermal expansion of the contents.

If welding is to be done on the outside surface of a vessel, and if the area is otherwise safe for the use of an open flame, the vessel need not be gas freed if one of the following procedures is employed:

a. In a vessel that is partly filled with liquid, welding may be done 3 feet or more below the level of the liquid if adequate precautions have been taken to prevent burning through the tank or vessel wall. Welding should not be conducted on a partly filled vessel which is double walled and/or which has an internal lining such as glass, plastic, or alloy until inspection and analysis are made to determine how to perform the work safely. (Important: see API RP 2201, which describes precautions and procedures for hot tapping and welding on a pressurized vessel. Also, see Sections 7 and 8 herein.)

b. Welding may be permitted if chemical analysis or other reliable evidence indicates that the petroleum-products vessel contains an atmosphere incapable of being ignited because it is too rich or too lean or is non-combustible or non-reactive and that adequate precautions have been taken to prevent burning through the vessel wall.

Note: If steam is used as a means of displacing oxygen, every part of the vessel shall be heated by the steam to a temperature of at least 77°C (170°F) so that the oxygen concentration is reduced to a point at which no ignitable mixture can be formed. In addition, when steam-inerting, pockets of oxygen or vapors may be present. Visible discharge of steam from an opening is insufficient evidence that the atmosphere within is not explosive. If carbon dioxide, nitrogen, or another inert gas is used as the displacing medium, it shall be introduced so that the atmosphere in all parts of the vessel is completely displaced. (Carbon dioxide should be introduced using a fixed-pipe system electrically bonded to the container to avoid a buildup of static electricity or a static-charged cloud). The relative weight of the gas compared with that of air must be taken into account. Steam or inert gas shall not be emitted from vessels or piping in the vicinity of hot work. The effectiveness of the purging must be determined by chemical analysis. Even if the oxygen content has been reduced to less than 10 percent by volume, it is not safe to assume that the vessel does not contain an ignitable mixture (for example, hydrogen's and carbon monoxide's limiting oxygen index is less than 10 percent). For more precise data, U.S. Bureau of Mines Bulletin 627 should be consulted. An ordinary combustible-gas detector can identify only the flammable characteristics of a combustible-gas/air mixture. Such a detector cannot accurately measure combustible-gas/inert-gas mixtures if the oxygen content is below certain levels. Check the manufacturer's recommendations for proper operation of these detectors under such conditions.

Hot work shall not be performed on one vessel or piece of equipment within a unit while other parts of the same unit are in operation unless it is determined that no unconfined flammable or combustible material is present in the work area and that no reasonable probability exists of any such material entering the area while hot work is being done. Precautions could include isolating the vessel or equipment from the remaining units, covering common drains, and/or keeping a steady flow of water on any adjacent leaks, flows, or drips that could result in flammable vapors.

Hot work shall not be performed on vessels under vacuum, unless an engineering evaluation is performed.

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SECTION 7—WORK IN BUILDINGS AND PITS

Precautions should be taken when hot work is to be done in buildings such as compressor rooms, manifold and pump houses, pits, receiving houses, and blending and container filling rooms while any equipment within these areas or buildings is in operation. If hot work is done within buildings or pits, the equipment to be worked on shall be isolated and freed from gas and oil. If necessary, other equipment shall be depressurized. The area shall be ventilated until gases or vapors are eliminated and appropriate toxic gas and/or combustible-gas tests show that the area is safe for hot work. Air movers, approved for service in hazardous locations, may be used to assist in ventilation. Confined space permit and safe work requirements shall be followed if applicable. Any combustible materials shall be protected or moved to another area if there is any chance of heat transfer or contact with sparks or hot slag.

SECTION 8—WORK ON PIPING

The need to free piping of flammable gases or liquids should be determined on the same basis used for vessels. The presence of flammable vapor can be established with a properly calibrated combustible-gas detector. Periodic retests or continuous monitoring may be necessary as the work proceeds. Other hazardous materials (for example, combustible liquids, corrosives, or toxics) may be present which could be harmful to personnel performing the work. Appropriate precautions must be taken to monitor exposures and to protect personnel.

An important precaution before welding or cutting piping is to properly isolate the piping, (for example, blinding, disconnecting, or double blocking and bleeding). The type of isolation used will depend on the contents, pressure, or piping configuration. A single valve may be used, depending on conditions *and if isolation is not required* (for example, welding on inert gas piping, water lines, units out of service, and so on). However, under the confined space standard (29 CFR 1910.146), a single closed valve is not approved as a means of isolation under OSHA. In addition, adequate provisions, such as venting, should be made to protect against overpressure due to thermal expansion of the contents. The actual determination that a pipe is safe, is a step-by-step procedure that must be followed closely prior to issuing a hot work permit.

Piping that has contained compressed air from a lubricated compressor may have an internal film of oil. Welding should not begin until the oil film has been removed or the pipe inerted.

Under certain conditions, it may be necessary to weld on or hot tap a pipe while the line is in operation. This work can be performed safely provided that qualified personnel take strict precautions both in preparation and in execution to avoid burning through the pipe wall. These precautions include determining that (a) the area is safe for an open flame, (b) the pipe wall is sound and sufficiently thick, (c) there is a minimum flow for heat dissipation, where required, and (d) the methods, tools, and equipment are safe and proper. Please note, however, that providing necessary technical information and recommended procedures for welding on equipment in service is beyond the scope of this publication (see API RP 2201 for more information on this subject).

API Publ 570 contains information on repairs, alterations, and rerating of in-service piping systems.

Preservative coatings are another important issue involving welding, cutting, and heating pipes and other surfaces. OSHA describes the precautions that should be taken for preservative coatings in 29 CFR 1926.354. The standard requires that when welding, cutting, or heating a surface with a preservative coating, a test shall be made by a competent person to determine the coating's flammability if the flammability is not already known. Further, OSHA states that when coatings are determined to be highly flammable, they shall be stripped from the area to be heated in order to prevent ignition. In enclosed spaces, all surfaces covered with toxic preservatives shall be stripped of all toxic coatings for a distance of at least 4 inches from the area of heat application, or the employees shall be protected by air line respirators. Also, the preservative coatings shall be removed a sufficient distance from the area to be heated to ensure that the temperature of the unstripped metal will not be appreciably raised. Artificial cooling of the metal surrounding the heating area may be used to limit the size of the area required to be cleaned.

SECTION 9—OTHER WELDING WORK

It is an accepted practice to repair certain leaks (such as on the Catalyst section of Fluid Catalytic Cracking Unit (FCCU)) on hydrocarbon lines, vessels, and other equipment while the unit remains in operation. Such repairs are accomplished by welding on patch plates and/or using a prefabricated enclosure.

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Small leaks can be safely repaired when proper precautions are taken. Larger leaks may require more elaborate procedures, such as the use of nitrogen to flush gas or liquids away from the leak through a line to a discharge location remote from the point of welding. The exact techniques vary considerably as there are many considerations involved, including the size and location of the leak, piping contents, the configuration of the equipment, operating pressures, and so forth.

It is essential that whatever repair procedures are used, appropriate precautions are taken to ensure the safety of all personnel involved, and that written procedures exist or are developed prior to performing such operations. A briefing to review the procedures should occur prior to the start of the job, and a hot work permit issued consistent with OSHA requirements, for example, 29 CFR 1910.119.

Factors to consider regarding the on-stream repair decision process and precautions to be taken include the following:

a. Compliance with company policy and permit procedures.
b. Needs for oxygen and gas tests.
c. Monitoring for toxic materials.
d. Size and location of the leak.
e. Location, length, and depth of the crack.
f. Whether the crack is in the parent metal, weld, heat-affected zone, or all three.
g. Metal thickness.
h. Size and proximity of weld or heat-affected zone.
i. Amount of possible thinning due to corrosion or erosion.
j. Needs for inerting.
k. Nature of PWHT requirements, if any.

Important: See API RP 2201 for more information.

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